



Seat No. : _____

7047

December-2008

Data Structures and Algorithms

FOR REFERENCE

Time : 3 Hours]

[Max. Marks : 100

(2 : 30 P.M. to 5 : 30 P.M.)

- Instructions :**
- (1) Attempt **all** questions from both sections.
 - (2) Use Separate Answer book for each sections.
 - (3) Figures to right indicate full marks.
 - (4) Assume suitable additional data, if necessary.

SECTION – I

1. Answer the following questions : 18
 - (a) What is difference between data type and data structure? What are the types of data structure? What role a data structure plays in problem solving ? 06
 - (b) What is Tower of Hanoi problem? Explain Tower of Hanoi problem for $n = 4$. 06
 - (c) Write an algorithm for inserting and deleting an element from Circular Queue. 06

2. Attempt the following questions : 16
 - (a) Write an algorithm **OR** C program for singly linked list which performs following operations : 08
 - (1) Insert a node such that linked list is in ascending order with respect to the information field.
 - (2) Delete a node after specified position.
 - OR**
 - (a) Write an algorithm **OR** C program for doubly linked list which performs following operations : 08
 - (1) Insert a node at the end of the linked list.
 - (2) Delete a node at specified position.
 - (b) Convert the following notation into postfix notation using stack : 03
 $(A + B) / (C * D * (E + F * (G^H^K))) * X + Y$
 - (c) Evaluate following expression using stack : 03
 $ABC^D * E / + F +$, Where $A = 2, B = 3, C = 2, D = 4, E = 1, F = 3$
 - (d) Differentiate following terms : 02
 - (1) Linear Search and Binary Search.
 - (2) Internal Sorting and External Sorting.

3. Answer the following questions : (Any Four)

16

- (a) Application of Stack
- (b) Priority Queue
- (c) Storage representation of 3-dimensional array
- (d) Sequential file and direct file organization
- (e) Multi key access organization

SECTION – II

4. Answer the following questions :

18

- (a) Write an algorithm for Radix sort method. Trace the algorithm on following data : 11, 60, 100, 20, 30, 80, 75, 65, 55 06
- (b) Write nonrecursive algorithm for pre order traversal of binary tree. 06
- (c) Explain following terms :(Any Four) 04
 - (a) graph
 - (b) forest
 - (c) critical node
 - (d) complete binary tree
 - (e) isolated node
- (d) Non linear data structure is more powerful, justify the statement. 02

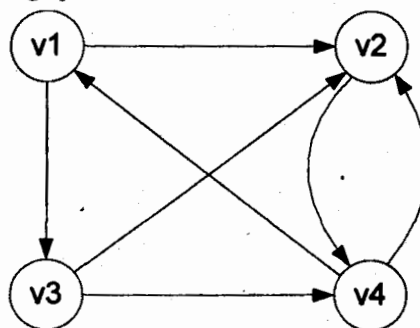
5. Attempt the following questions :

16

- (a) Write an algorithm for quick sort and also sort the following data using it : 42, 23, 13, 65, 58, 94, 36, 99, 87 07

OR

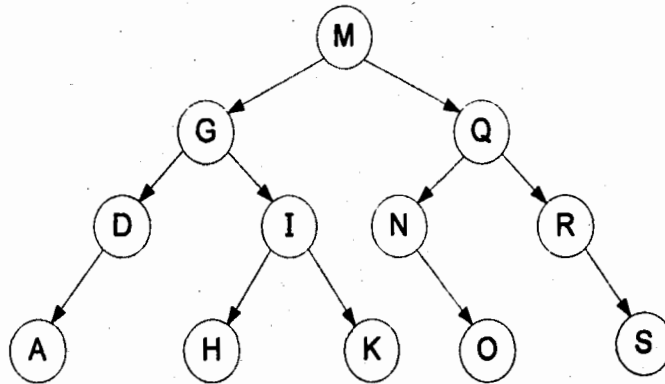
- (a) What is sparse matrix ? Give its sequential and linked representation. Give its advantages and disadvantages of both representations. 07
- (b) Answer the following questions with reference to the given tree : 06



- (1) Write an adjacency matrix A for the graph.
- (2) Find A^2 from the adjacency matrix A.
- (3) What is the in degree, out degree and total degree of the v1 node ?

OR

- (b) (1) Construct a 4-way tree with the following data : 04
 80, 120, 150, 90, 113, 126, 135, 142, 68, 73, 202
- (2) Construct a binary search tree with the following data : 02
 5, 3, 1, 5, 6, 7, 11, 9, 10, 14, 12, 15, 13
- (c) Answer the following questions with reference to the given tree : 03



- (1) Find in order traversal for the tree.
- (2) Find post order traversal for the tree.
- (3) Find pre order traversal for the tree.

OR

- (c) State following statements are true or false. Give appropriate reason 03
- (1) Tree is recursive structure.
 - (2) There is no disadvantage of buddy system.
 - (3) Bubble sort is more efficient than selection sort.

6. Answer the following : (Any **Four**) 16
- (a) Threaded Binary tree
 - (b) Hashing Techniques (Any Three)
 - (c) BFS and DPS in graph
 - (d) Weight Balanced tree
 - (e) Explain the storage representation techniques of graph